COLLEGE ATTENDANCE MANAGEMENT SYSTEM

DBMS MINI PROJECT - ONE PAGE REPORT

Project title	College Attendance Management System	**Database**	SQLite (sqlite3)
Frontend	HTML, Bootstrap, Jinja2	**Backend**	Python 3, Flask
Tools	VS Code, GitHub	**Project type**	Database-driven web application

INTRODUCTION AND PROBLEM STATEMENT

The system stores students, subjects and daily attendance in a relational database. Staff can manage records, mark attendance, and quickly identify students whose attendance is below 75%.

OBJECTIVE

Build a backend and frontend supporting Create, Read, Update and Delete (CRUD). Apply SQL joins, views, a trigger and transactions, and generate attendance reports.

DATABASE DESIGN

Table	Columns and constraints
Departments	dept_id PK; dept_name UNIQUE
Students	student_id PK; roll_no and email UNIQUE; dept_id FK; year CHECK 1-4
Subjects	subject_id PK; dept_id FK; year CHECK 1-4
Attendance	att_id PK; student_id and subject_id FKs; UNIQUE(student, subject, date); status CHECK Present/Absent; cascade deletes
Attendance log	log_id PK; att_id, old_status, new_status, changed_at; populated by trigger

IMPLEMENTATION

Backend (Flask): admin login with session, student and subject CRUD, attendance marking, dashboard counts, and reports. Frontend: Bootstrap pages for login, dashboard, students, subjects, attendance checkbox sheet, and reports. SQL features: attendance_summary and low_attendance_view (<75%), relational constraints, audit trigger, and transaction-based daily attendance.

RELATIONSHIPS

Departments 1:N students; departments 1:N subjects; students 1:N attendance; subjects 1:N attendance. Attendance connects students and subjects by date.

CONCLUSION AND FUTURE SCOPE

The project demonstrates normalization, constraints, joins, views and transaction management. Future extensions include teacher and student logins, charts, PDF export and automatic low-attendance alerts.

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